

TrES-3b

R-2660

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150602 · created 2026-08-30T00:10:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457175.7735 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.052
Transit depth (Rp/Rs)^2	3.68 +/- 2.0 [%]
Orbital Inclination (inc)	81.31 +/- 2.18
Airmass coefficient 1 (a1)	0.926 +/- 0.02
Airmass coefficient 2 (a2)	0.062 +/- 0.029
Scatter in the residuals of the lightcurve fit is	2.2 %
Best Comparison Star	#3 - [365, 454]
Optimal Method	PSF photometry
Transit Duration (day)	0.048 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.192 ± 0.052 vs 0.16309 (deviation 0.56σ)
Duration fit vs published	0.048 d vs 0.0567 d (deviation 0.36σ)
Mid-time O–C	2.1 min
Depth SNR	3.7
Residual scatter	2.2 %

Light curve

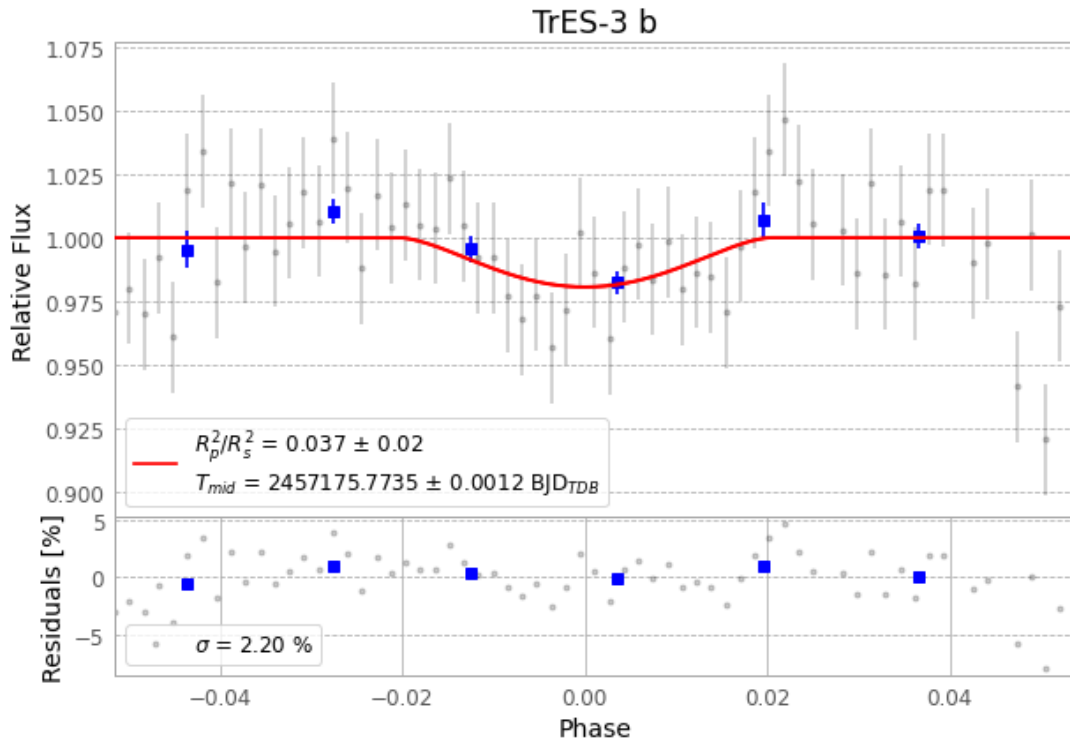


Figure 1 — FinalLightCurve_TrES-3 b_2015-06-01.png (SHA-256 e13c40fdecf53f1f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [123, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9b3c85683febe073...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3150602045112.FITS → TRES-3150602080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150602.log (ea856f5e32b5...), FinalLightCurve_TrES-3 b_2015-06-01.png (e13c40fdecf5...), FinalLightCurve_TrES-3 b_2015-06-01.csv (84e789c4fbcf...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:10Z.